Real Gases: Vander Waal's Equation

Vander Waal's Equation (VDW) for *n* mole real gas: $\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$

Let us consider two hypothetical cases to show the size effect and attraction effect on the Pr. of the gas.

(a) For the real gas, $a = 0$ (i.e, no intermolecular attraction exist) but $b \neq 0$ (size is considered).

VDW Eqn.: $\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$, but $a = 0$,

So, $P = \frac{nRT}{V - nb} > P_{id}$ Since, $P_{id} = \frac{nRT}{V}$

It means that the molecular size (repulsive attraction) creates higher Pr. than that observed by the ideal gas where molecules have no vol.

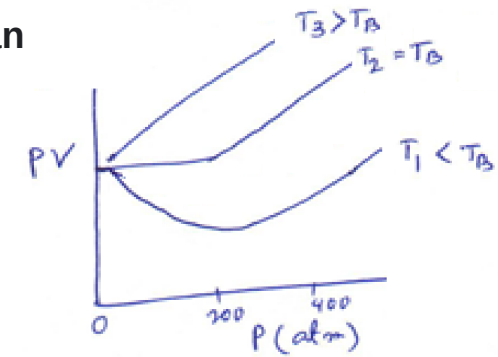
(b) For the real gas $a \neq 0$ (intermolecular attraction exists) but $b = 0$, i.e, no size of the molecule.

The VDW Eqn.: $P = \frac{nRT}{V} - \frac{an^2}{V^2} < P_{id}$, since $b = 0$ and $P_{id} = \frac{nRT}{V}$

Thus, intermolecular attraction effect reduces the Pr. of the real gas.

Real Gases: Calculation of Boyle Temp., T_B

✓ $T_B \rightarrow$ **The temperature at which real gases obey ideal gas equation over an appreciable pressure range**



VDW Equn: $P = \dfrac{RT}{V-b} - \dfrac{a}{V^2}$

$$PV = \frac{RTV}{V-b} - \frac{a}{V}$$

Thus, $\left[\dfrac{\partial(PV)}{\partial P}\right]_T = \left[\dfrac{\partial(PV)}{\partial V}\right]_T \times \left[\dfrac{\partial V}{\partial P}\right]_T$

$$\equiv \left[-\frac{RTb}{(V-b)^2} + \frac{a}{V^2}\right] \times \left(\frac{\partial V}{\partial P}\right)_T$$

at T_B, $\left[\dfrac{\partial(PV)}{\partial P}\right]_T = 0$ when $P \rightarrow 0$

when $T = T_B$, $\left[\dfrac{\partial(PV)}{\partial P}\right]_T = 0$ but $\left(\dfrac{\partial V}{\partial P}\right)_T \neq 0$

Hence, $\dfrac{RT_B b}{(V-b)^2} = \dfrac{a}{V^2}$

$$\Rightarrow T_B = \frac{a}{Rb}\left(\frac{V-b}{V}\right)^2$$

Since, $P \rightarrow 0$, V is large and $\dfrac{V-b}{V} \approx 1$

Therefore, $T_B = \dfrac{a}{Rb}$ ← **This is the expression of Boyle temp. for a gas obeying VDW Eq.**

Question: Calculate *a*/*b* for a gas for which T_b = 227 °C?

Real Gases: Explanation of Amagat's Curves in the Light of VDW Eq.

VDW Equn.. for 1 mole real gas:

$$\left(P + \frac{a}{V^2}\right)(V-b) = RT$$

$$\Rightarrow PV - Pb + \frac{a}{V} - \frac{ab}{V^2} = RT$$

$$\Rightarrow PV = RT + Pb - \frac{a}{V}$$ [Neglecting the small term $\frac{ab}{V^2}$]

$$\Rightarrow \frac{PV}{RT} = 1 + \frac{Pb}{RT} - \frac{a}{VRT},$$ Since $Z = \frac{PV}{RT}$

$$\Rightarrow Z = 1 + \frac{1}{RT}\left(Pb - \frac{a}{V}\right)$$

$$= 1 + \frac{1}{RT}\left(Pb - \frac{aP}{RT}\right)$$

$$= 1 + \frac{P}{RT}\left(b - \frac{a}{RT}\right)$$

$PV = RT$

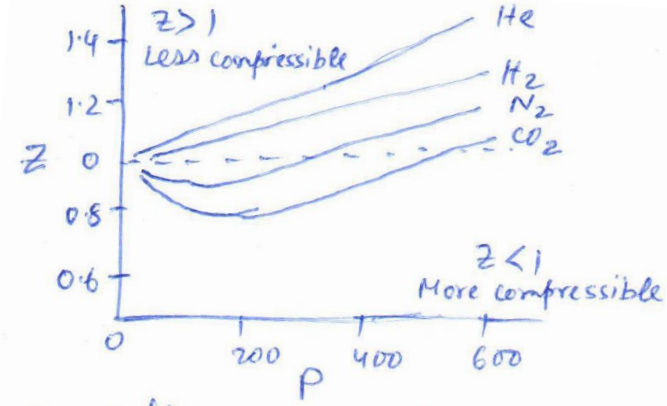
$aPV = aRT$

$\frac{aP}{RT} = \frac{a}{V}$

→ **This shows that Z = f (T, P)**

✓ **This Eq. can be used to explain Amagat's curves quantitatively at low P to moderate P region**

Real Gases: Explanation of Amagat's Curves in the Light of VDW Eq.



For diff. gases at const. T.

$$Z = 1 + \frac{P}{RT}\left(b - \frac{a}{RT}\right)$$

$a_{CO_2} = 3.95$ atm L^2 mol^{-2}

$b_{CO_2} = 0.04$ L mol^{-1}

CO_2: "*a*" is high as we have seen that the gas is easily liquefiable

Thus, $\frac{a}{RT} > b$ in the equn. and $b - \frac{a}{RT} = -ve$.

Intermolecular attraction dominates over the size effect.

Thus, the slope of Z vs. P curve is −ve for CO_2 at moderate Pr. region. This shows that the value of Z decreases with increase of P and it is also found in the curve.

For H_2 gas, 'a' is small as it is not easily liquified.

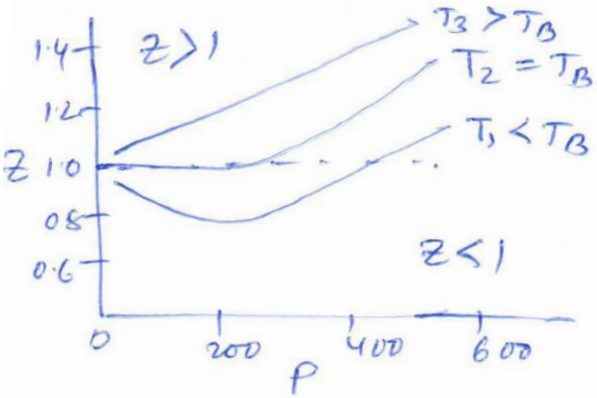
$\frac{a}{RT} < b$ and the slope of Z vs. P curve for H_2 is +ve.

The value of Z increases with increase of P.

$a_{H_2} = 0.22$ atm L^2 mol^{-2}

$b_{H_2} = 0.02$ L mol^{-1}

Real Gases: Explanation of Amagat's Curves in the Light of VDW Eq.



For CO_2 at diff. T

$$Z = 1 + \frac{P}{RT}\left(b - \frac{a}{RT}\right)$$

(i) when $T < T_B$, $T < \frac{a}{Rb}$, or $b < \frac{a}{RT}$, thus $b - \frac{a}{RT} = -ve$

It means that when $T < T_B$, the value of Z decreases with increase of P at the moderate Pr. The effect is due to intermolecular attraction dominates over the size effect. So, for CO_2, $Z < 1$ and the gas is more compressible.

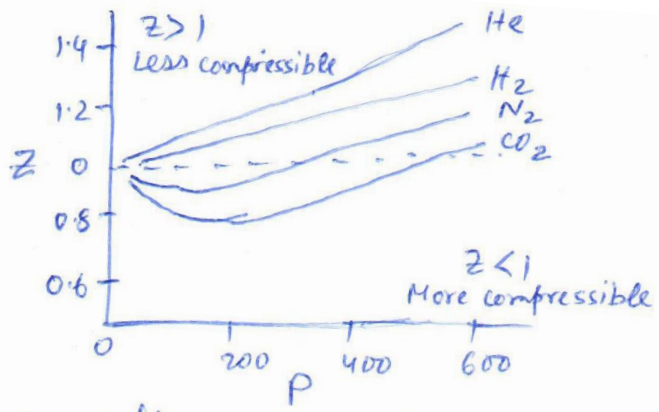
(ii) when $T = T_B = \frac{a}{Rb}$ or $b = \frac{a}{RT}$, or $b - \frac{a}{RT} = 0$, So, $Z = 1$.

The gas shows ideal behavior. The size effect compensates the effect due to intermolecular attraction of the gas. Z runs parallel to P-axis up to certain range of Pr. at low Pr. region.

(iii) when, $T > T_B$ means $T > \frac{a}{Rb}$, or $b > \frac{a}{RT}$.

Thus, $b - \frac{a}{RT} = +ve$. Thus, Z increases with increase of P when $T > T_B$. The size effect dominates over the effect due to intermolecular attraction and $Z > 1$ and the gas is less compressible.

Real Gases: Explanation of Amagat's Curves in the Light of VDW Eq.



For diff. gases at const. T.

(*) For H_2 and He, 0°C is greater than their T_B values and so Z vs. P slope becomes +ve.

$$Z = 1 + \frac{P}{RT}\left(b - \frac{a}{RT}\right)$$

(*) At very low P ($P \to 0$) and at high T, volume is very large and both the size effect and attraction effect becomes negligible. Thus Pb and $\frac{aP}{RT}$ are negligibly small and $Z = 1$. The gas behaves ideal.

Critical Phenomena – Andrews' Curve

A gas can be liquefied by lowering of temp. and increasing Pr.
But influence of temp. is more important.

↓

Most gases are liquefied at ordinary Pr. by suitably lowering of temp. But, a gas can not be liquefied unless its temp. is below a certain value depending on the nature of the gas.
→ Critical temp. (T_c)

- ✓ **Above T_c, the gas can not be liquefied whatever high Pr. may be applied to.**

↓

A gas can only be liquefied when the temp. is kept below T_c of the gas. The Pr. required to liquefy the gas at its T_c is called critical Pr. (P_c) and the vol. occupied by 1 mole at T_c and P_c is called critical vol. (V_c).

↓

- ✓ **T. Andrews (1869), in his experiment with 1 mole CO_2 collected data of *P* vs. *V* at various temps., to obtain critical constants values by drawing *P* vs. *V* at different temps.**

- ✓ **These curves are known as Andrews curves.**

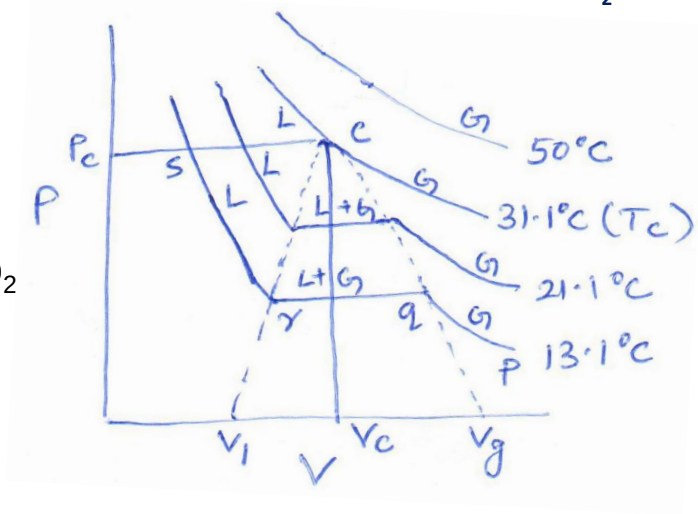
Ideal Gas

Critical Phenomena – Andrews' Curve

- ✓ Let us discuss the isotherm for 1 mole CO_2 at 13.1 ^oC (below T_c), ***pqrs***.
- ✓ The point ***p*** represents the gaseous CO_2 at low Pr.
- ✓ As Pr. increased, V is correspondingly decreased (Boyle's law).
- ✓ At point ***q***, the gaseous CO_2 begins to liquify and the Pr. at the point is the saturation vapor Pr. of CO_2. As the vol. is decreased, more of the gaseous CO_2 transforms into liquid CO_2 but Pr. remains unchanged.
- ✓ This isothermal conversion continues up to ***r*** when all the gaseous CO_2 is converted into liquid CO_2.

Now the curve ***rs*** is very steep as the liquid is highly incompressible.

Andrews isotherm for 1 mole CO_2



When the temp. of 21.1 ^oC is taken for the study, similar curve is obtained except the liquid begins to form at higher saturation Pr. and the range of vol. over which condensation occurs is smaller.

At 31.1 ^oC, the plateau shrinks to a point and this temp. is the critical temp. (T_c) of CO_2.

When the temp. is increased to 50 ^oC, the isotherm approaches more closely to that of ideal gas; no plateau is observed and no liquid is formed.

The dotted line encloses a dome-shaped area within which liquid and gas are co-existent. The highest point C of the area indicates the critical point. On the right side of the area, gas alone is present and at the left side liquid.

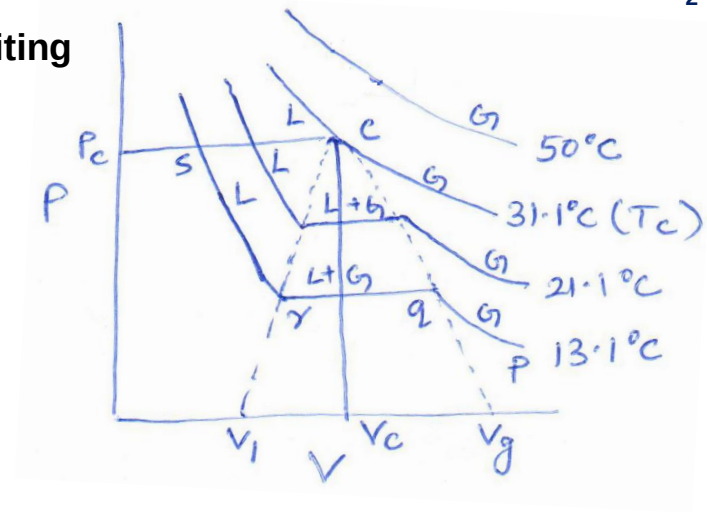
For CO_2; T_c = **31.1 ^oC**
P_c = **72.9 atm**
V_c = **94.2 cc/mol**

Critical Phenomena – Condition of the Critical Point (C)

The critical point is the limiting point of a series of horizontal two-phase lines. So, the slope of the horizontal lines as well as the limiting point (C) is;

$$\left(\frac{\partial P}{\partial V}\right)_T = 0$$

Andrews isotherm for 1 mole CO_2



- **Along the critical temp. isotherm, the slope is zero at the critical point (C) and is –ve on either side of the point. Thus, the slope is maximum (zero value is greater than –ve values) at the critical point.**

- **This slope is function of V and its derivative w.r.t. V is again zero at the point. So,**

$$\left[\frac{\partial}{\partial V}\left(\frac{\partial P}{\partial V}\right)_T\right]_T = \left(\frac{\partial^2 P}{\partial V^2}\right)_T = 0 \text{ at the point.}$$

Thus, the condition of the critical point is given by

$$\left(\frac{\partial P}{\partial V}\right)_T = 0 \quad \text{and} \quad \left(\frac{\partial^2 P}{\partial V^2}\right)_T = 0$$

That means, both slope and curvature at the point C are zero